Linear algebra in R

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1 Introduction

This note has two goal: 1) Introducing linear algebra (vectors and matrices) and 2) showing how to work with these concepts in R.

2 Vectors

2.1 Vectors

A column vector is a list of numbers stacked on top of each other, e.g.

$$a = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$$

A row vector is a list of numbers written one after the other, e.g.

$$b = (2, 1, 3)$$

In both cases, the list is ordered, i.e.

$$(2, 1, 3) \neq (1, 2, 3).$$

We make the following convention:

- In what follows all vectors are column vectors unless otherwise stated.
- However, writing column vectors takes up more space than row vectors. Therefore we shall frequently write vectors as row vectors, but with the understanding that it really is a column vector.

A general n–vector has the form

$$a = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$$

where the a_is are numbers, and this vector shall be written $a = (a_1, \ldots, a_n)$.
A graphical representation of 2–vectors is shown Figure 1. Note that row and

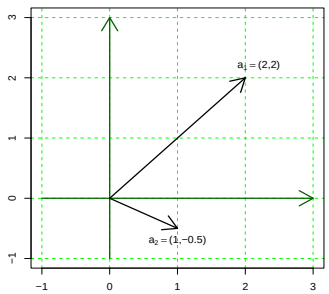


Figure 1: Two 2-vectors

column vectors are drawn the same way.

```
> a <- c(1, 3, 2)
> a
```

```
[1] 1 3 2
```

The vector `a` is in R printed "in row format" but can really be regarded as a column vector, cfr. the convention above.

2.2 Transpose of vectors

Transposing a vector means turning a column (row) vector into a row (column) vector. The transpose is denoted by "$^\top$".

Example 1

$$\begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}^\top = [1,3,2] \quad \texttt{og} \quad [1,3,2]^\top = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}$$

□

Hence transposing twice takes us back to where we started:

$$a = (a^\top)^\top$$

```
> t(a)
```

```
     [,1] [,2] [,3]
[1,]    1    3    2
```

2.3 Multiplying a vector by a number

If a is a vector and α is a number then αa is the vector

$$\alpha a = \begin{bmatrix} \alpha a_1 \\ \alpha a_2 \\ \vdots \\ \alpha a_n \end{bmatrix}$$

See Figure 2.

Example 2

$$7 \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 7 \\ 21 \\ 14 \end{bmatrix}$$

□

```
> 7 * a
```

```
[1]  7 21 14
```

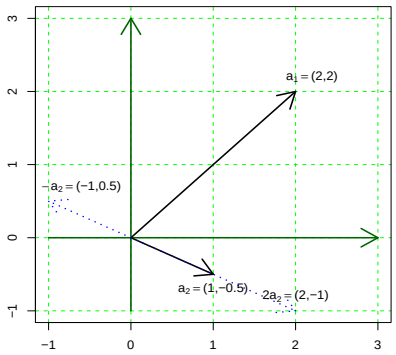


Figure 2: Multiplication of a vector by a number

2.4 Sum of vectors

Let a and b be n–vectors. The sum $a + b$ is the n–vector

$$a + b = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} + \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} = \begin{bmatrix} a_1 + b_1 \\ a_2 + b_2 \\ \vdots \\ a_n + b_n \end{bmatrix} = b + a$$

See Figure 3 and 4. Only vectors of the same dimension can be added.

Example 3

$$\begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} + \begin{bmatrix} 2 \\ 8 \\ 9 \end{bmatrix} = \begin{bmatrix} 1+2 \\ 3+8 \\ 2+9 \end{bmatrix} = \begin{bmatrix} 3 \\ 11 \\ 11 \end{bmatrix}$$

□

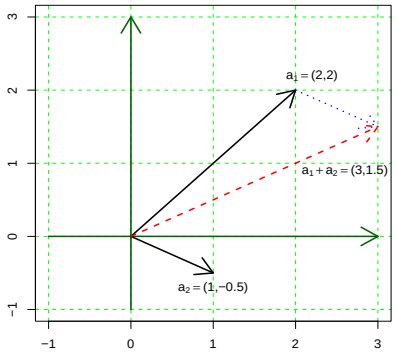


Figure 3: Addition of vectors

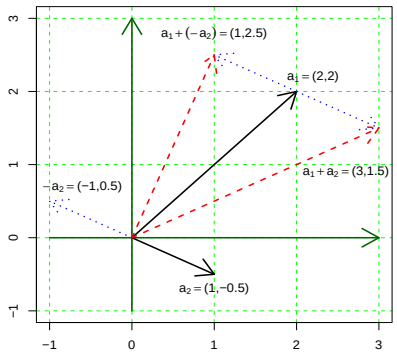


Figure 4: Addition of vectors and multiplication by a number

```
> a <- c(1, 3, 2)
> b <- c(2, 8, 9)
> a + b
```

```
[1]  3 11 11
```

2.5 (Inner) product of vectors

Let $a = (a_1, \ldots, a_n)$ and $b = (b_1, \ldots, b_n)$. The (inner) product of a and b is

$$a \cdot b = a_1 b_1 + \cdots + a_n b_n$$

Note, that the product is a number – not a vector.

```
> sum(a * b)
```

```
[1] 44
```

2.6 The length (norm) of a vector

The length (or norm) of a vector a is

$$||a|| = \sqrt{a \cdot a} = \sqrt{\sum_{i=1}^{n} a_i^2}$$

```
> sqrt(sum(a * a))
```

```
[1] 3.741657
```

2.7 The 0–vector and 1–vector

The 0-vector (1–vector) is a vector with 0 (1) on all entries. The 0–vector (1–vector) is frequently written simply as 0 (1) or as 0_n (1_n) to emphasize that its length n.

```
> rep(0, 5)
```

```
[1] 0 0 0 0 0
```

```
> rep(1, 5)
```

```
[1] 1 1 1 1 1
```

2.8 Orthogonal (perpendicular) vectors

Two vectors v_1 and v_2 are orthogonal if their inner product is zero, written

$$v_1 \perp v_2 \Leftrightarrow v_1 \cdot v_2 = 0$$

```
> v1 <- c(1, 1)
> v2 <- c(-1, 1)
> sum(v1 * v2)
```

```
[1] 0
```

3 Matrices

3.1 Matrices

An $r \times c$ matrix A (reads "an r times c matrix") is a table with r $r$ows og c $c$olumns

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1c} \\ a_{21} & a_{22} & \dots & a_{2c} \\ \vdots & \vdots & \ddots & \vdots \\ a_{r1} & a_{r2} & \dots & a_{rc} \end{bmatrix}$$

Note that one can regard A as consisting of c columns vectors put after each other:

$$A = [a_1 : a_2 : \cdots : a_c]$$

```
> A <- matrix(c(1, 3, 2, 2, 8, 9), ncol = 3)
> A
```

```
     [,1] [,2] [,3]
[1,]    1    2    8
[2,]    3    2    9
```

Note that the numbers $1, 3, 2, 2, 8, 9$ are read into the matrix column–by–column. To get the numbers read in row–by–row do

```
> A2 <- matrix(c(1, 3, 2, 2, 8, 9), ncol = 3, byrow = T)
> A2
```

```
     [,1] [,2] [,3]
[1,]    1    3    2
[2,]    2    8    9
```

3.2 Multiplying a matrix with a number

For a number α and a matrix A, the product αA is the matrix obtained by multiplying each element in A by α.

Example 4

$$7 \begin{bmatrix} 1 & 2 \\ 3 & 8 \\ 2 & 9 \end{bmatrix} = \begin{bmatrix} 7 & 14 \\ 21 & 56 \\ 14 & 63 \end{bmatrix}$$

□

```
> 7 * A
```

```
     [,1] [,2] [,3]
[1,]    7   14   56
[2,]   21   14   63
```

3.3 Transpose of matrices

A matrix is transposed by interchanging rows and columns and is denoted by "$\top$".

Example 5

$$\begin{bmatrix} 1 & 2 \\ 3 & 8 \\ 2 & 9 \end{bmatrix}^{\top} = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 8 & 9 \end{bmatrix}$$

□

Note that if A is an $r \times c$ matrix then $A^{\top}$ is a $c \times r$ matrix.

```
> t(A)
```

```
     [,1] [,2]
[1,]    1    3
[2,]    2    2
[3,]    8    9
```

3.4 Sum of matrices

Let A and B be $r \times c$ matrices. The sum $A + B$ is the $r \times c$ matrix obtained by adding A and B elementwise.

Only matrices with the same dimensions can be added.

Example 6

$$\begin{bmatrix} 1 & 2 \\ 3 & 8 \\ 2 & 9 \end{bmatrix} + \begin{bmatrix} 5 & 4 \\ 8 & 2 \\ 3 & 7 \end{bmatrix} = \begin{bmatrix} 6 & 6 \\ 11 & 10 \\ 5 & 16 \end{bmatrix}$$

□

```
> B <- matrix(c(5, 8, 3, 4, 2, 7), ncol = 3, byrow = T)
> A + B
```

```
     [,1] [,2] [,3]
[1,]    6   10   11
[2,]    7    4   16
```

3.5 Multiplication of a matrix and a vector

Let A be an $r \times c$ matrix and let b be a c-dimensional column vector. The product Ab is the $r \times 1$ matrix

$$Ab = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1c} \\ a_{21} & a_{22} & \dots & a_{2c} \\ \vdots & \vdots & \ddots & \vdots \\ a_{r1} & a_{r2} & \dots & a_{rc} \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_c \end{bmatrix} = \begin{bmatrix} a_{11}b_1 + a_{12}b_2 + \cdots + a_{1c}b_c \\ a_{21}b_1 + a_{22}b_2 + \cdots + a_{2c}b_c \\ \vdots \\ a_{r1}b_1 + a_{r2}b_2 + \cdots + a_{rc}b_c \end{bmatrix}$$

Example 7

$$\begin{bmatrix} 1 & 2 \\ 3 & 8 \\ 2 & 9 \end{bmatrix} \begin{bmatrix} 5 \\ 8 \end{bmatrix} = \begin{bmatrix} 1 \cdot 5 + 2 \cdot 8 \\ 3 \cdot 5 + 8 \cdot 8 \\ 2 \cdot 5 + 9 \cdot 8 \end{bmatrix} = \begin{bmatrix} 21 \\ 79 \\ 82 \end{bmatrix}$$

□

```
> A %*% a
```

```
     [,1]
[1,]   23
[2,]   27
```

Note the difference to

```
> A * a
```

```
     [,1] [,2] [,3]
[1,]    1    4   24
[2,]    9    2   18
```

Figure out yourself what goes on!

3.6 Multiplication of matrices

Let A be an $r \times c$ matrix and B a $c \times t$ matrix, i.e. $B = [b_1 : b_2 : \cdots : b_t]$. The product AB is the $r \times t$ matrix given by:

$$AB = A[b_1 : b_2 : \cdots : b_t] = [Ab_1 : Ab_2 : \cdots : Ab_t]$$

Example 8

$$\begin{bmatrix} 1 & 2 \\ 3 & 8 \\ 2 & 9 \end{bmatrix} \begin{bmatrix} 5 & 4 \\ 8 & 2 \end{bmatrix} = \left[\begin{bmatrix} 1 & 2 \\ 3 & 8 \\ 2 & 9 \end{bmatrix} \begin{bmatrix} 5 \\ 8 \end{bmatrix} : \begin{bmatrix} 1 & 2 \\ 3 & 8 \\ 2 & 9 \end{bmatrix} \begin{bmatrix} 4 \\ 2 \end{bmatrix} \right]$$

$$= \begin{bmatrix} 1 \cdot 5 + 2 \cdot 8 & 1 \cdot 4 + 2 \cdot 2 \\ 3 \cdot 5 + 8 \cdot 8 & 3 \cdot 4 + 8 \cdot 2 \\ 2 \cdot 5 + 9 \cdot 8 & 2 \cdot 4 + 9 \cdot 2 \end{bmatrix} = \begin{bmatrix} 21 & 8 \\ 79 & 28 \\ 82 & 26 \end{bmatrix}$$

□

Note that the product AB can only be formed if the number of rows in B and the number of columns in A are the same. In that case, A and B are said to be conforme.

In general AB and BA are not identical.

A MNEMONIC FOR MATRIX MULTIPLICATION is :

$$\begin{bmatrix} 1 & 2 \\ 3 & 8 \\ 2 & 9 \end{bmatrix} \begin{bmatrix} 5 & 4 \\ 8 & 2 \end{bmatrix} = \begin{array}{cc|cc} & & 5 & 4 \\ & & 8 & 2 \\ \hline 1 & 2 & 1 \cdot 5 + 2 \cdot 8 & 1 \cdot 4 + 2 \cdot 2 \\ 3 & 8 & 3 \cdot 5 + 8 \cdot 8 & 3 \cdot 4 + 8 \cdot 2 \\ 2 & 9 & 2 \cdot 5 + 9 \cdot 8 & 2 \cdot 4 + 9 \cdot 2 \end{array} = \begin{bmatrix} 21 & 8 \\ 79 & 28 \\ 82 & 26 \end{bmatrix}$$

```
> A <- matrix(c(1, 3, 2, 2, 8, 9), ncol = 2)
> B <- matrix(c(5, 8, 4, 2), ncol = 2)
> A %*% B
```

```
     [,1] [,2]
[1,]   21    8
[2,]   79   28
[3,]   82   26
```

3.7 Vectors as matrices

One can regard a column vector of length r as an $r \times 1$ matrix and a row vector of length c as a $1 \times c$ matrix.

3.8 Some special matrices

- An $n \times n$ matrix is a SQUARE MATRIX
- A matrix A is SYMMETRIC if $A = A^\top$.
- A matrix with 0 on all entries is the 0–MATRIX and is often written simply as 0.

- A matrix consisting of 1s in all entries is of written J.
- A square matrix with 0 on all off–diagonal entries and elements $d_1, d_2, \ldots, d_n$ on the diagonal a DIAGONAL MATRIX and is often written $diag\{d_1, d_2, \ldots, d_n\}$
- A diagonal matrix with 1s on the diagonal is called the IDENTITY MATRIX and is denoted I. The identity matrix satisfies that $IA = AI = A$.

- 0-matrix and 1-matrix

```
> matrix(0, nrow = 2, ncol = 3)
```

```
     [,1] [,2] [,3]
[1,]    0    0    0
[2,]    0    0    0
```

```
> matrix(1, nrow = 2, ncol = 3)
```

```
     [,1] [,2] [,3]
[1,]    1    1    1
[2,]    1    1    1
```

- Diagonal matrix and identity matrix

```
> diag(c(1, 2, 3))
```

```
     [,1] [,2] [,3]
[1,]    1    0    0
[2,]    0    2    0
[3,]    0    0    3
```

```
> diag(1, 3)
```

```
     [,1] [,2] [,3]
[1,]    1    0    0
[2,]    0    1    0
[3,]    0    0    1
```

Note what happens when `diag` is applied to a matrix:

```
> diag(diag(c(1, 2, 3)))
```

```
[1] 1 2 3
```

```
> diag(A)
```

```
[1] 1 8
```

3.9 Inverse of matrices

In general, the inverse of an $n \times n$ matrix A is the matrix B (which is also $n \times n$) which when multiplied with A gives the identity matrix I. That is,

$$AB = BA = I.$$

One says that B is A's inverse and writes $B = A^{-1}$. Likewise, A is Bs inverse.

Example 9 Let

$$A = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \quad B = \begin{bmatrix} -2 & 1.5 \\ 1 & -0.5 \end{bmatrix}$$

Now $AB = BA = I$ so $B = A^{-1}$. □

Example 10 If A is a 1×1 matrix, i.e. a number, for example $A = 4$, then $A^{-1} = 1/4$. □

Some facts about inverse matrices are:

- Only square matrices can have an inverse, but not all square matrices have an inverse.
- When the inverse exists, it is unique.
- Finding the inverse of a large matrix A is numerically complicated (but computers do it for us).

In Section **??** the issue of matrix inversion is discussed in more detail.

Finding the inverse of a matrix in R is done using the `solve()` function:

```
> A <- matrix(c(1, 3, 2, 4), ncol = 2, byrow = T)
> A
```

```
     [,1] [,2]
[1,]    1    3
[2,]    2    4
```

```
> B <- solve(A)
> B
```

```
     [,1] [,2]
[1,]   -2  1.5
[2,]    1 -0.5
```

```
> A %*% B
```

```
     [,1] [,2]
[1,]    1    0
[2,]    0    1
```

3.10 Solving systems of linear equations

Example 11 Matrices are closely related to systems of linear equations. Consider the two equations

$$\begin{aligned} x_1 + 3x_2 &= 7 \\ 2x_1 + 4x_2 &= 10 \end{aligned}$$

The system can be written in matrix form

$$\begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 7 \\ 10 \end{bmatrix} \quad \text{i.e. } Ax = b$$

Since $A^{-1}A = I$ and since $Ix = x$ we have

$$x = A^{-1}b = \begin{bmatrix} -2 & 1.5 \\ 1 & -0.5 \end{bmatrix} \begin{bmatrix} 7 \\ 10 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

A geometrical approach to solving these equations is as follows: Isolate x_2 in the equations:

$$x_2 = \frac{7}{3} - \frac{1}{3}x_1 \quad x_2 = \frac{1}{0}4 - \frac{2}{4}x_1$$

These two lines are shown in Figure 5 from which it can be seen that the solution is $x_1 = 1, x_2 = 2$.

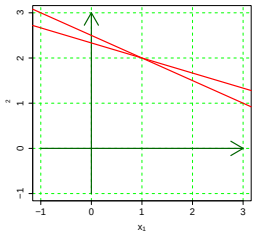


Figure 5: Solving two equations with two unknowns.

From the Figure it follows that there are 3 possible cases of solutions to the system

1. Exactly one solution – when the lines intersect in one point
2. No solutions – when the lines are parallel but not identical
3. Infinitely many solutions – when the lines coincide.

□

```
> A <- matrix(c(1, 2, 3, 4), ncol = 2)
> b <- c(7, 10)
> x <- solve(A) %*% b
> x
```

```
     [,1]
[1,]    1
[2,]    2
```

3.11 Trace

Missing

3.12 Determinant

Missing

3.13 Some additional rules for matrix operations

For matrices A, B and C whose dimension match appropriately: the following rules apply

$$(A+B)^\top = A^\top + B^\top$$
$$(AB)^\top = B^\top A^\top$$
$$A(B+C) = AB + AC$$
$$AB = AC \not\Rightarrow B = C$$

In genereal $AB \neq BA$

$$AI = IA = A$$

If α is a number then $\alpha AB = A(\alpha B)$

3.14 Details on inverse matrices*

3.14.1 Inverse of a 2×2 matrix*

It is easy find the inverse for a 2×2 matrix. When

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

then the inverse is

$$A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

under the assumption that $ab-bc \neq 0$. The number $ab-bc$ is called the determinant of A, sometimes written $|A|$. If $|A| = 0$, then A has no inverse.

3.14.2 Inverse of diagonal matrices*

Finding the inverse of a diagonal matrix is easy: Let

$$A = diag(a_1, a_2, \ldots, a_n)$$

where all $a_i \neq 0$. Then the inverse is

$$A^{-1} = diag(\frac{1}{a_1}, \frac{1}{a_2}, \ldots, \frac{1}{a_n})$$

If one $a_i = 0$ then A^{-1} does not exist.

3.14.3 Generalized inverse*

Not all square matrices have an inverse. However all square matrices have an infinite number of generalized inverses. A generalized inverse of a square matrix A is a matrix A^- satisfying that

$$AA^-A = A.$$

For many practical problems it suffice to find a generalized inverse.

3.14.4 Inverting an $n \times n$ matrix*

In the following we will illustrate one frequently applied methopd for matrix inversion. The method is called Gauss–Seidels method and many computer programs, including `solve()` use variants of the method for finding the inverse of an $n \times n$ matrix.
Consider the matrix A:

```
> A <- matrix(c(2, 2, 3, 3, 5, 9, 5, 6, 7), ncol = 3)
> A
```

```
     [,1] [,2] [,3]
[1,]    2    3    5
[2,]    2    5    6
[3,]    3    9    7
```

We want to find the matrix $B = A^{-1}$. To start, we append to A the identity matrix and call the result AB:

```
> AB <- cbind(A, diag(c(1, 1, 1)))
> AB
```

```
     [,1] [,2] [,3] [,4] [,5] [,6]
[1,]    2    3    5    1    0    0
[2,]    2    5    6    0    1    0
[3,]    3    9    7    0    0    1
```

On a matrix we allow ourselves to do the following three operations (sometimes called elementary operations) as often as we want:

1. Multiply a row by a (non–zero) constant.
2. Multiply a row by a (non–zero) constant and add the result to another row.
3. Interchange two rows.

The aim is to perform such operations on `AB` in a way such that one ends up with a 3×6 matrix which has the identity matrix in the three leftmost columns. The three rightmost columns will then contain $B = A^{-1}$.
Recall that writing e.g. `AB[1,]` extracts the enire first row of AB.

- First, we make sure that `AB[1,1]=1`. Then we subtract a constant times the first row from the second to obtain that `AB[2,1]=0`, and similarly for the third row:

  ```
  > AB[1, ] <- AB[1, ]/AB[1, 1]
  > AB[2, ] <- AB[2, ] - 2 * AB[1, ]
  > AB[3, ] <- AB[3, ] - 3 * AB[1, ]
  > AB
  ```

  ```
       [,1] [,2] [,3] [,4] [,5] [,6]
  [1,]    1  1.5  2.5  0.5    0    0
  [2,]    0  2.0  1.0 -1.0    1    0
  [3,]    0  4.5 -0.5 -1.5    0    1
  ```

- Next we ensure that `AB[2,2]=1`. Afterwards we subtract a constant times the second row from the third to obtain that `AB[3,2]=0`:

```
> AB[2, ] <- AB[2, ]/AB[2, 2]
> AB[3, ] <- AB[3, ] - 4.5 * AB[2, ]
```

- Now we rescale the third row such that `AB[3,3]=1`:

```
> AB[3, ] <- AB[3, ]/AB[3, 3]
> AB
```

```
     [,1] [,2] [,3]       [,4]      [,5]       [,6]
[1,]    1  1.5  2.5  0.5000000 0.0000000  0.0000000
[2,]    0  1.0  0.5 -0.5000000 0.5000000  0.0000000
[3,]    0  0.0  1.0 -0.2727273 0.8181818 -0.3636364
```

  Then `AB` has zeros below the main diagonal.

- We then work our way up to obtain that `AB` has zeros above the main diagonal:

```
> AB[2, ] <- AB[2, ] - 0.5 * AB[3, ]
> AB[1, ] <- AB[1, ] - 2.5 * AB[3, ]
> AB
```

```
     [,1] [,2] [,3]       [,4]        [,5]       [,6]
[1,]    1  1.5    0  1.1818182 -2.04545455  0.9090909
[2,]    0  1.0    0 -0.3636364  0.09090909  0.1818182
[3,]    0  0.0    1 -0.2727273  0.81818182 -0.3636364
```

```
> AB[1, ] <- AB[1, ] - 1.5 * AB[2, ]
> AB
```

```
     [,1] [,2] [,3]       [,4]        [,5]       [,6]
[1,]    1    0    0  1.7272727 -2.18181818  0.6363636
[2,]    0    1    0 -0.3636364  0.09090909  0.1818182
[3,]    0    0    1 -0.2727273  0.81818182 -0.3636364
```

Now we extract the three rightmost columns of `AB` into the matrix `B`. We claim that `B` is the inverse of `A`, and this can be verified by a simple matrix multiplication

```
> B <- AB[, 4:6]
> A %*% B
```

```
              [,1]          [,2]          [,3]
[1,]  1.000000e+00 3.330669e-16 1.110223e-16
[2,] -4.440892e-16 1.000000e+00 2.220446e-16
[3,] -2.220446e-16 9.992007e-16 1.000000e+00
```

So, apart from rounding errors, the product is the identity matrix, and hence $B = A^{-1}$. This example illustrates that numerical precision and rounding errors is an important issue when making computer programs.

4 Least squares

Consider the table of pairs (x_i, y_i) below.

x	1.00	2.00	3.00	4.00	5.00
y	3.70	4.20	4.90	5.70	6.00

A plot of y_i against x_i is shown in Figure 6.

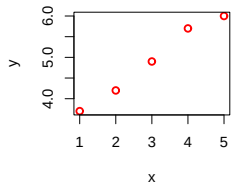


Figure 6: Regression

The plot in Figure 6 suggests an approximately linear relationship between y and x, i.e.

$$y_i = \beta_0 + \beta_1 x_i \text{ for } i = 1, \ldots, 5$$

Writing this in matrix form gives

$$y = \begin{bmatrix} y_1 \\ y_2 \\ \ldots \\ y_5 \end{bmatrix} \approx \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_5 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \boldsymbol{X\beta}$$

The first question is: Can we find a vector β such that $y = X\beta$? The answer is clearly no, because that would require the points to lie exactly on a straight line.
A more modest question is: Can we find a vector $\hat{\beta}$ such that $X\hat{\beta}$ is in a sense "as close to y as possible". The answer is yes. The task is to find $\hat{\beta}$ such that the length of the vector

$$e = y - X\beta$$

is as small as possible. The solution is

$$\hat{\beta} = (X^\top X)^{-1} X^\top y$$

```
> y
```

```
[1] 3.7 4.2 4.9 5.7 6.0
```

```
> X
```

```
     x
[1,] 1 1
[2,] 1 2
[3,] 1 3
[4,] 1 4
[5,] 1 5
```

```
> beta.hat <- solve(t(X) %*% X) %*% t(X) %*% y
> beta.hat
```

```
    [,1]
   3.07
x 0.61
```

5 A neat little exercise – from a bird's perspective

On a sunny day, two tables are standing in an English country garden. On each table birds of unknown species are sitting having the time of their lives.
A bird from the first table says to those on the second table: "Hi – if one of you come to our table then there will be the same number of us on each table". "Yeah, right", says a bird from the second table, "but if one of you comes to our table, then we will be twice as many on our table as on yours".
Question: How many birds are on each table? More specifically,

- Write up two equations with two unknowns.
- Solve these equations using the methods you have learned from linear algebra.
- Simply finding the solution by trial–and–error is considered cheating.